

The statistics of soaring flight

Displacement scaling across terrain, circuit types and wing classes

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OFFSITE
2026

EConophysics Lab

CFM

ILB Institut
Louis
Bachelier

The soaring flight cycle

2



Transition

Glide, lose height



Climb

Circle, regain height

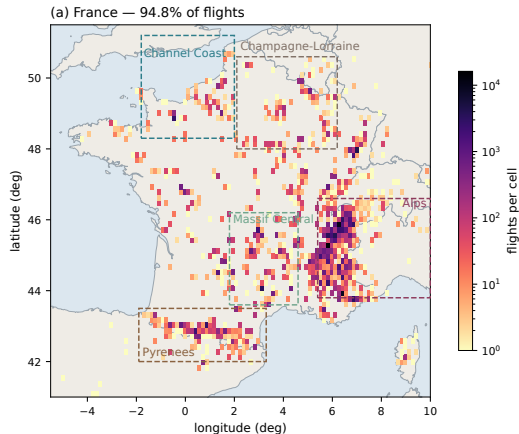


Search

Find rising air



Thermals let pilots cover long distances without an engine



FFVL distance competition

186,052

Paraglider tracklogs
1999-2026

6,716

Hang-glider tracklogs
2001-2026

Regional boxes

- Channel Coast
- Champagne-Lorraine
- Massif Central
- Alps
- Pyrenees

IGC trajectories + metadata: time, position, altitude, site, circuit and wing class

Mean-square displacement (MSD)

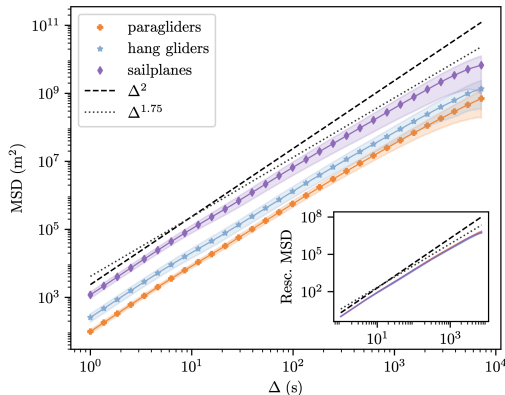
- 1 Along each flight f , of duration T_f

$$m_f(\tau) = \frac{1}{T_f - \tau} \int_0^{T_f - \tau} |\mathbf{x}_f(t + \tau) - \mathbf{x}_f(t)|^2 dt$$

- 2 Over the N flights: $M_2(\tau) = \frac{1}{N} \sum_f m_f(\tau)$

$$M_2(\tau) \propto \tau^{2H}: \quad H = \frac{1}{2} \text{ diffusive, } H = 1 \text{ ballistic}$$

Vilpellet et al.: French flights, 2016–2021



Same $H \approx 0.88$ for paragliders, hang gliders and sailplanes: a universal exponent

My internship: what lies behind this universality?

From raw tracklogs to the fixed cohort C_{10000}

5

Preprocessing

Clean the recorded flight



Selection and coordinates



Uniform time grid



Savitzky–Golay smoothing



Retained (of downloads)

156,406

paragliders, 84%

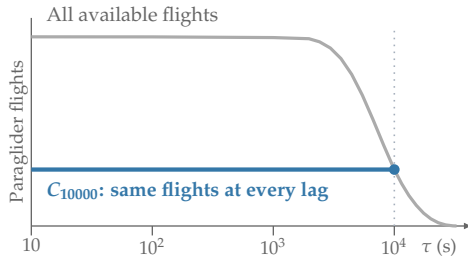
6,094

hang gliders, 91%



Fixed cohort C_{10000}

Segments of at least 12,500 s (≈ 3.5 h)



In C_{10000} (of retained)

46,273

paragliders, 30%

2,326

hang gliders, 38%

Which flight characteristics shape MSD and H ?

6

Circuit

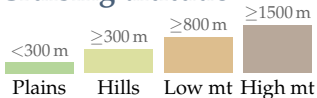


open



closed

Cruising altitude



Wing class

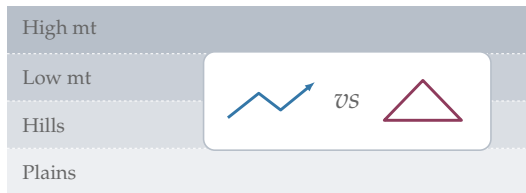
Beginners EN A/B/C
Experts EN D/CCC

Weather

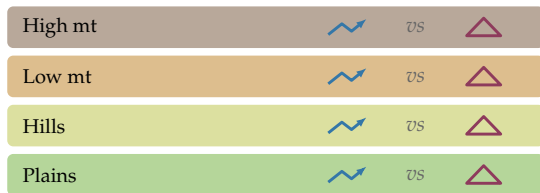


not observed
at flight scale

Marginal across all altitude classes



Stratified within each altitude class

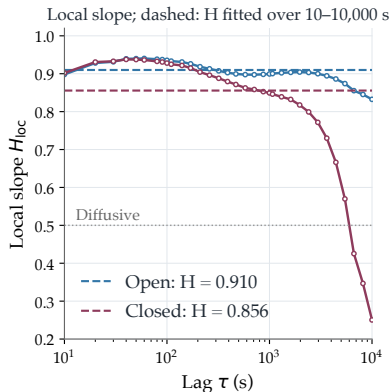
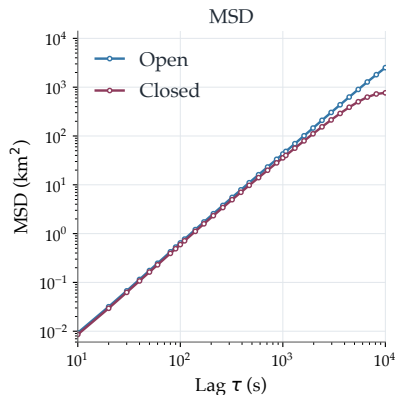


First compare groups overall, then within the same altitude class.

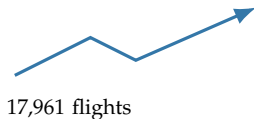
Each flight is assigned to an altitude class by z_0 , the GNSS altitude of its first fix after trimming.

Open vs closed: how do MSD and H differ?

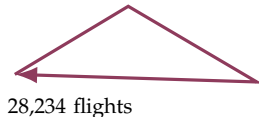
7



Open circuits



Closed circuits



$$H_{\text{open}} - H_{\text{closed}} = 0.054 [0.053, 0.056] > 0 \quad (90\% \text{ paired interval})$$

Slower closed-route MSD growth is consistent with turns and returns.

78 unclassified flights excluded. Nominal 90% cluster-bootstrap CIs; bands likely too narrow (Appendix).

Does the open–closed H gap change with altitude?

8



Open: higher H in lowlands

- **Lowlands:** no deviations for lift
- **Mountains:** deviations to known lift

Closed: higher H in mountains

- **Lowlands:** turn back any time
- **Mountains:** way home follows ridges

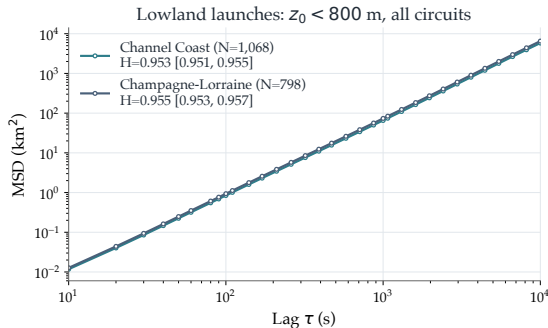
H less reliable: no clean power law over all decades (slide 7)

Open > closed everywhere: gap 0.13 in lowlands, 0.02–0.05 in mountains.

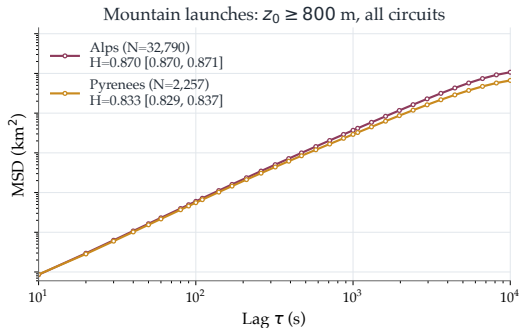
Mountain terrain constrains both circuits, so the gap narrows.

How does H differ by region?

9



Lowlands: almost equal
Channel Coast – Champagne-Lorraine
–0.002 [–0.005, 0.001]



Mountains: Alps above Pyrenees
Alps – Pyrenees
0.037 [0.033, 0.041]

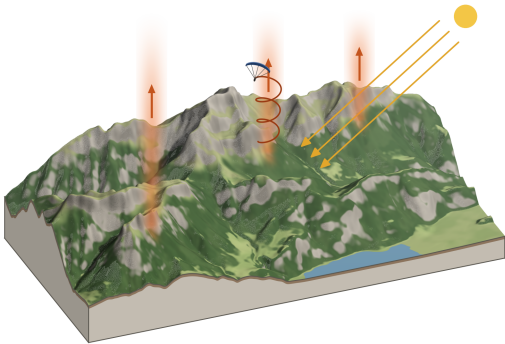
H separates the two mountain ranges more clearly than the two plains.

Why does H separate the ranges but not the plains?

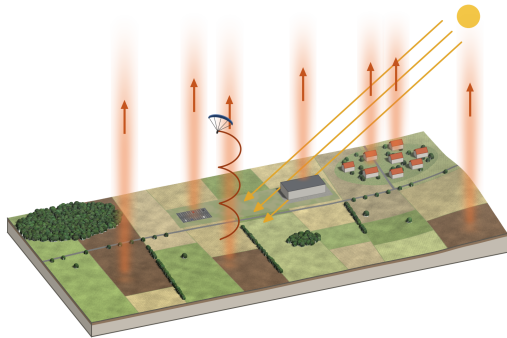
10

Mountains: sunlit rocky ridges

Surface colour, material and moisture control local heating.



Lowlands: scattered triggers



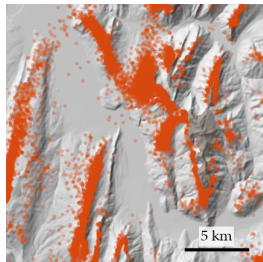
Rocky slopes anchor mountain lift. Lowland sources form scattered patches.

Different relief may give different H , while similar lowland triggers may give similar H .

Mountain lift follows ridges, lowland lift spreads aloft

11

Alps

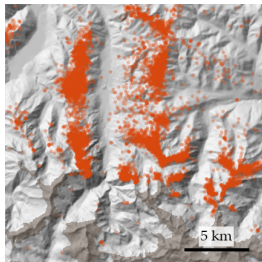


$H = 800$ m

49,105 climb crossings

[Open GIF](#)

Pyrenees

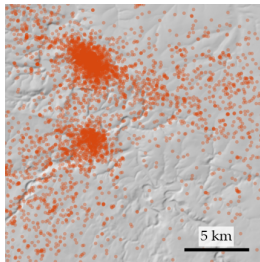


$H = 800$ m

10,557 climb crossings

[Open GIF](#)

Channel Coast

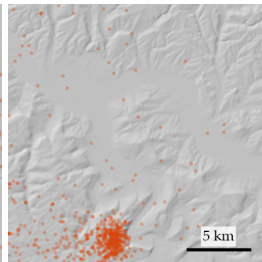


$H = 800$ m

4,140 climb crossings

[Open GIF](#)

Champagne-Lorraine



$H = 800$ m

660 climb crossings

[Open GIF](#)

● climb crossing

■ terrain above the plane

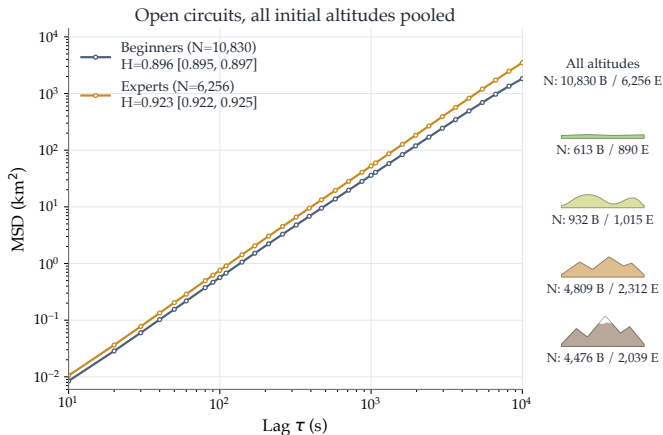
[Cell locations \(Appendix\)](#)

Higher up, a random location may offer easier access to thermals in lowlands.

This may explain similar Hurst exponents across lowlands and differences between mountain ranges.

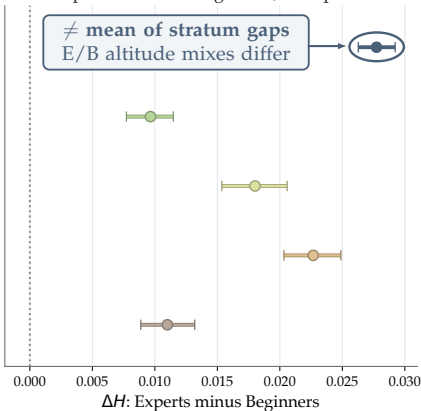
Open circuits: expert–beginner H gap by altitude

12



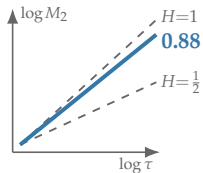
Open circuits. B: Beginners; E: Experts

$\neq$ mean of stratum gaps
E/B altitude mixes differ



Vilpellet et al. also found $H_{\text{expert}} > H_{\text{beginner}}$,
with experts spending less time searching for thermals.

Scaling



Strongly superdiffusive

Universal across paragliders, hang gliders and sailplanes

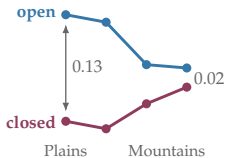
Route and wing



Open routes spread faster

Experts spread faster

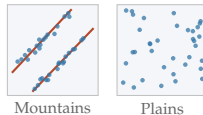
Terrain



Mountains narrow the gap

- Open-route $H \downarrow$
- Closed-route $H \uparrow$

Lift geography



Mountains: ridge climbs

Plains: climbs spread out

Strongly superdiffusive everywhere; route and terrain set how strongly.

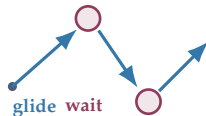
Segmentation



Revise Vilpellet's HMM

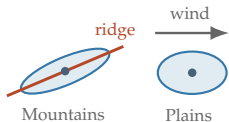


Stochastic model



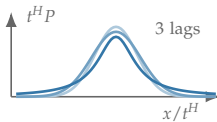
Which model fits best?

Anisotropy



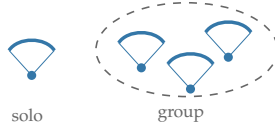
Ridge axis or wind

Distribution scaling



Do distributions collapse?

Solo vs group flight



Does group flight change H ?

Thank you
for your attention!

Appendix: clean the recorded flight

↔ slide 5 16

1 GNSS altitude

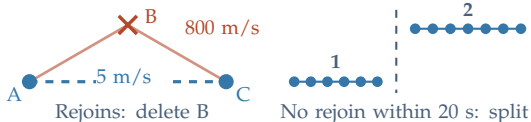
GNSS height → Trajectory $z(t)$

Barometer → Cleaning checks

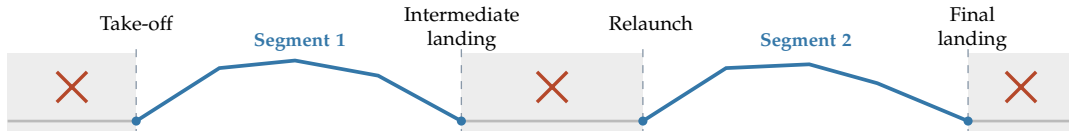
GNSS: absolute height without weather drift.
Same high-frequency noise as the barometer.

2 Fix-level cleaning

One fix = (t, φ, λ, z) : time, latitude, longitude, height
Delete an unreachable fix only if the track rejoins without it; otherwise split the flight.



3 Airborne interval



Trim before take-off and after landing.
Stops are not labeled: we infer them from position and altitude.

4 Flight selection

Minimum flight thresholds

Time ≥ 40 min

1.13% excluded ($N = 191,108$)

Path ≥ 20 km

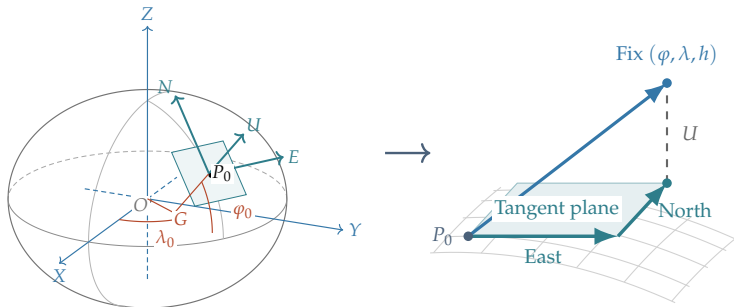
0.09% excluded ($N = 188,942$)

Altitude range ≥ 600 m

2.68% excluded ($N = 188,768$)

5 Coordinates

Latitude, longitude, height $\rightarrow$ Earth-centred (X, Y, Z)
 $\rightarrow$ local east-north distances in metres



Earth model: WGS84 ellipsoid (GPS reference)

P_0 : first trimmed fix; fixed for the flight. $z = h_{\text{GNSS}}$, not U .

6a Gaps: bridge or split

Gap g between consecutive fixes:

- $g \leq g_{\max}$: bridge it.
- $g > g_{\max}$: split into segments.

$$g_{\max} = \min(10\Delta t, \max(20 \text{ s}, 2\Delta t))$$

Example: $\Delta t = 5 \text{ s}$, $g_{\max} = 20 \text{ s}$

Short gap
15 s: bridge



Long gap
30 s: split



● Raw fix
(jittered)

● Grid point

○ Flagged as
reconstructed

6b Time grid: jitter removal

Per segment: grid $t_k = t_0 + k\Delta t$, native Δt .

- Interpolate E , N , z at every t_k .
- No fix within $\Delta t/2$ of t_k : flag it as reconstructed.



Keep a segment only if it lasts $\geq 90 \text{ s}$ and $\leq 10\%$ of its grid is reconstructed.

No segment kept: flight dropped. 20,798 paragliders, 193 hang gliders: the largest loss.

7 Savitzky-Golay fit

Fit a cubic to E , N , z in a moving window, within each segment.

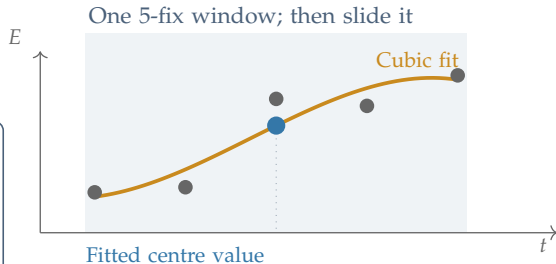
Window: odd, ≥ 5 fixes.

Velocity and acceleration: derivatives of the fitted cubic

$$v_i = P'_i(0), \quad a_i = P''_i(0), \quad u = t - t_i$$

Noise std, for an independent error σ per fix:

	velocity	acceleration
Fitted cubic	$0.95 \sigma / \Delta t$	$0.53 \sigma / \Delta t^2$
Fix-to-fix differences	$1.41 \sigma / \Delta t$	$2.45 \sigma / \Delta t^2$



Retained flights after all seven stages (share of downloaded logs)

156,406 **paragliders** 84.1%

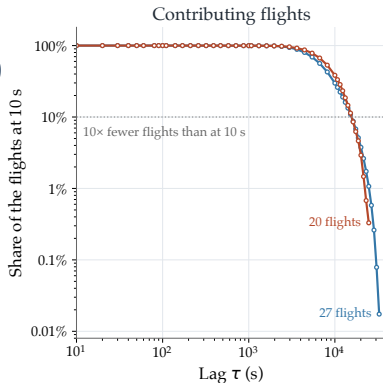
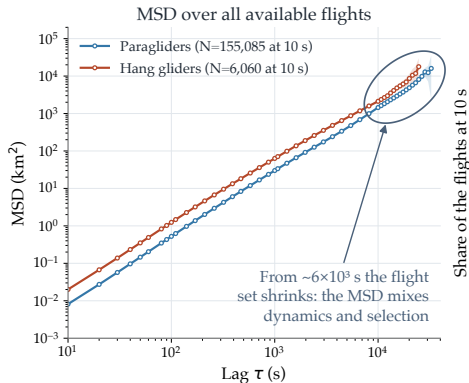
6,094 **hang gliders** 90.7%

Schematic; the cubic fit is exact for the synthetic fixes shown.

Appendix: which flights contribute at each lag?

↔ slide 5 20

All available flights: a flight enters at lag τ if one of its segments has $\tau \leq 0.8 T_s$ ($\geq 1/5$ of it left as origins for m_f).



Desiderata

- 1 The same flights at every lag
- 2 The longest possible lag range
- 3 $N \approx$ of the same order as at 10 s

At long lags only long flights remain, and their number falls steeply.

Any lag-dependent statistic then mixes dynamics with a changing set of flights.

Appendix: can we afford a fixed population?

↔ slide 5 21

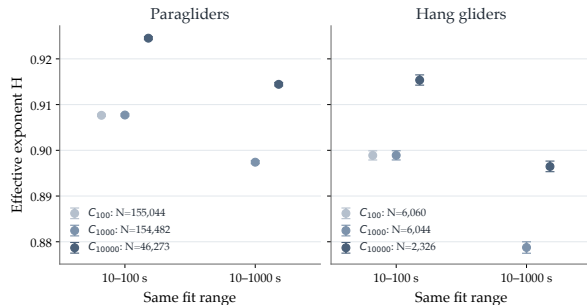
C_{10000} : select once for the largest lag

$\tau_{\max} = 10,000 \text{ s} \leq 0.8 T_s$: $T_s \geq 12,500 \text{ s}$
Only these segments enter, at every lag.

Advantages of C_{10000}

- 1 Same flights and segments, 10 to 10,000 s, with no loss of flights beyond $6 \times 10^3 \text{ s}$
- 2 Three decades of lag; at $\tau_{\max}$ each segment still offers at least 251 origins for m_f
- 3 Sample still large, same order as at 10 s:
46,273 paragliders (30%)
2,326 hang gliders (38%)

Cost of selecting longer flights



Nested cohorts, identical fit ranges, paired bootstrap draws.

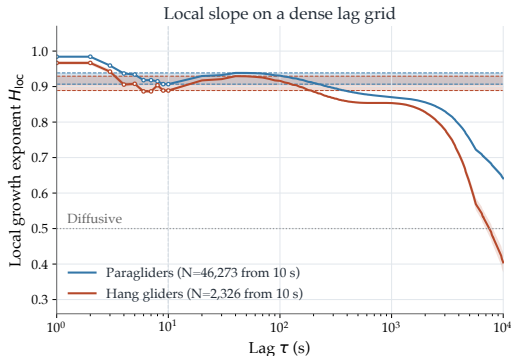
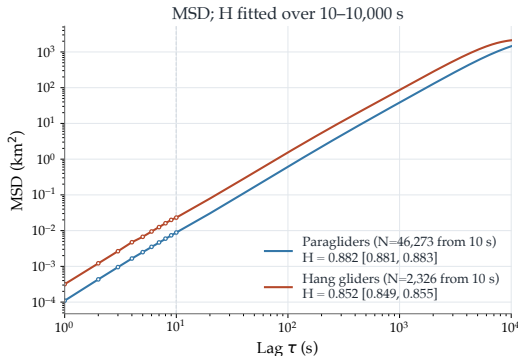
Long flights raise H by 0.016–0.018 on matched lag ranges; origins still decrease with lag.
About 2% in H buys the same flights at every lag over three decades.

Displacement growth from 1 s to 10,000 s

22

C_{10000} : changing support below 10 s; the same flights at every lag from 10 to 10,000 s.

$M_2(\tau) \propto \tau^{2H}$: $H = \frac{1}{2}$ diffusion $H = 1$ ballistic motion



$$H_{\text{PG}} = 0.882 [0.881, 0.883] > H_{\text{HG}} = 0.852 [0.849, 0.855] \text{ over } 10\text{--}10,000 \text{ s}$$

Nominal 90% bootstrap CIs; clusters sampled with replacement. Clustering remains too fine; bands likely too narrow.

1 Group flights

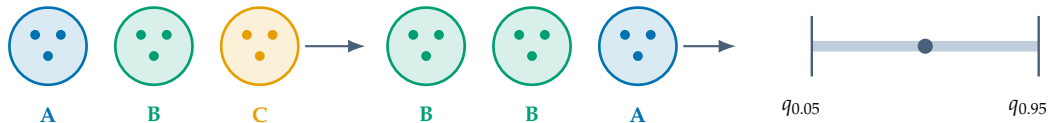
Take-off site $\times$ calendar day.
Aim: independent groups.

2 Resample groups

Sample with replacement.
Keep each group together.

3 Recompute

1,000 replicates.
Take the 5th–95th percentiles.



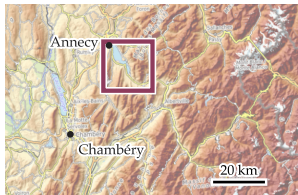
Current clusters are too small to absorb all shared flight conditions.

Intervals likely underestimate uncertainty; clustering is still being refined

Appendix: locations of the 20 km cells

↔ slide 11 24

Alps: Lake Annecy



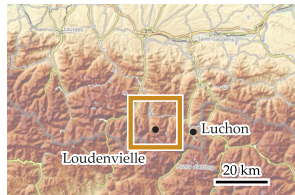
Centre (WGS84)

45.82374° N
 6.22085° E

Bounds (Lambert-93, m)

E 940000–960000
N 6520000–6540000

Pyrenees: Louron Valley



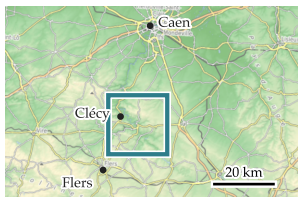
Centre (WGS84)

42.77884° N
 0.43624° E

Bounds (Lambert-93, m)

E 480000–500000
N 6180000–6200000

Channel Coast: Suisse Normande



Centre (WGS84)

48.88143° N
 0.40990° W

Bounds (Lambert-93, m)

E 440000–460000
N 6860000–6880000

Champagne-Lorraine: Mouzon



Centre (WGS84)

49.63141° N
 5.07544° E

Bounds (Lambert-93, m)

E 840000–860000
N 6940000–6960000

Appendix: why the pooled gap is larger

↔ slide 12 25

Same gap within each class

Toy values; ordinary arithmetic means, not fitted H

	Beginners	Experts	Gap
Lowlands	0.86	0.90	0.04
Mountains	0.76	0.80	0.04

10 equally weighted flights per group

Experts ● ● ● ● ● ● ● ● ● ● → **0.89**
 $(9 \times 0.90 + 1 \times 0.80) / 10$

Beginners ● ● ● ● ● ● ● ● ● ● → **0.77**
 $(1 \times 0.86 + 9 \times 0.76) / 10$

$$0.89 - 0.77 = 0.12 > 0.04$$

Slide 12: the observed mix

Open circuits; lowlands = Plains + Hills



● Lowlands ● Mountains

Experts include more lowland flights, where H is higher for both groups.

Different altitude mixes can enlarge the pooled gap.

For H : pool MSD curves, then fit. Do not average the four gaps.

Flight archive

FFVL Coupe Fédérale de Distance: IGC tracks, dates, declared sites, circuits and wing classes.

parapente.ffvl.fr/cfd/liste delta.ffvl.fr/cfd/liste

Maps and terrain

[IGN RGE ALTI](#): terrain elevation of the 20 km cells, 25 m grid.

[OpenTopoMap](#): appendix cell maps (OSM / SRTM, CC-BY-SA).

geo.api.gouv.fr: commune names on the appendix maps.

Analysis code

Code and figure scripts, with the download URL of each geographic source.